Stage 1 Hard Drill — With Solutions

Chapter 3.1 — Sampling Mechanics & Finite Population Selection

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Hypergeometric PMF, Basic Draw

A prime-brokerage desk has $N = 25$ active counterparties, $K = 6$ flagged as "toxic flow." You interact with $n = 4$ of them (no repeats). Find $\mathbb{P}$(exactly 2 toxic).

Solution.

$$\mathbb{P}(X=2) = \frac{\binom{6}{2}\binom{19}{2}}{\binom{25}{4}} = \frac{15 \cdot 171}{12650} = \frac{2565}{12650} = \frac{513}{2530} \approx \boxed{0.2028}$$

⋆ *Intuition:* split the choice into "2 from the toxic pool" × "2 from the clean pool," normalize by the unrestricted way to pick 4 from 25 — the entire method in one line, every time.

Q2 — Complementary Counting Under Audit

A bond lot has $N = 40$ issues, $K = 5$ below investment grade. Compliance samples $n = 8$ without replacement. Find $\mathbb{P}$(at least 2 sub-grade).

Solution. Complement is faster than summing $k = 2, \ldots, 5$ directly:

$$\mathbb{P}(X=0) = \frac{\binom{35}{8}}{\binom{40}{8}} = \frac{25172}{82251} \approx 0.3060$$

$$\mathbb{P}(X=1) = \frac{\binom{5}{1}\binom{35}{7}}{\binom{40}{8}} = \frac{35960}{82251} \approx 0.4372$$

$$\mathbb{P}(X \geq 2) = 1 - 0.3060 - 0.4372 = \boxed{0.2568}$$

⋆ *Intuition:* "at least 2" almost always means "1 minus the first two PMF terms," not a 4-term sum — the complement rule from Example 2 generalizes to any small threshold, not just "at least 1."

Q3 — Mean & Variance in One Pass

A portfolio audit: $N = 60$ positions, $K = 15$ high-beta names, sample $n = 12$ without replacement. Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution.

$$\mathbb{E}[X] = n\frac{K}{N} = 12 \cdot \frac{15}{60} = \boxed{3}$$

$$\mathrm{Var}(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1} = 12(0.25)(0.75) \cdot \frac{48}{59} = \boxed{\frac{108}{59} \approx 1.831}$$

⋆ *Speed trick:* compute the "naive" Binomial variance $np(1-p) = 2.25$ first, then multiply by the FPC $\frac{N-n}{N-1} = \frac{48}{59} \approx 0.8136$ as a separate last step — never fold the FPC into the middle of the arithmetic, it's easy to drop a factor that way.

Q4 — Working Backwards: Recover N

A desk reports: "We sampled $n = 10$ trades without replacement from a blotter that is exactly half-flagged ($K/N = 0.5$). The reported variance of the flagged-count is exactly 1.6." Recover N (and K).

Solution. With $p = K/N = 0.5$:

$$\mathrm{Var}(X) = np(1-p) \cdot \frac{N-n}{N-1} = 10(0.25) \cdot \frac{N-10}{N-1} = 2.5 \cdot \frac{N-10}{N-1} = 1.6$$

Cross-multiply: $2.5(N-10) = 1.6(N-1) \Rightarrow 2.5N - 25 = 1.6N - 1.6 \Rightarrow 0.9N = 23.4$

$$N = \boxed{26}, \qquad K = 0.5N = \boxed{13}$$

⋆ *Intuition:* this is the same equation as every forward FPC problem, just solved for N instead of $\mathrm{Var}(X)$ — recognizing "solve the identity backwards" is itself a Stage 1 skill, not a different formula.

Q5 — How Good Is the Binomial Approximation, Exactly?

$N = 800$ resting quotes, $K = 160$ are stale ($p = 0.2$), sample $n = 15$. Compare exact $\mathbb{P}(X = 3)$ to the Binomial$(15, 0.2)$ approximation, and give the relative error.

Solution. Exact:

$$\mathbb{P}(X=3) = \frac{\binom{160}{3}\binom{640}{12}}{\binom{800}{15}} \approx \boxed{0.2525}$$

Binomial approx: $\binom{15}{3}(0.2)^3(0.8)^{12} \approx \boxed{0.2501}$

Relative error $\approx \dfrac{0.2525 - 0.2501}{0.2501} \approx \boxed{0.95\%}$

⋆ *Intuition:* here $n/N = 15/800 \approx 1.9\%$, safely under the 5% rule of thumb — and the actual error (under 1%) confirms the rule is, if anything, conservative. On a Stage 1 MCQ with large N, skip straight to Binomial and don't look back.

Q6 — The Duality Trick, By Necessity

$N = 500$ accounts, only $K = 3$ flagged for review. A sample of $n = 120$ trades is audited. Find $\mathbb{P}$(exactly 1 flagged account appears) — and do it by hand.

Solution. Direct form needs $\binom{497}{119}$ — infeasible by hand. Use the duality $\mathrm{Hypergeometric}(N, K, n) \overset{d}{=} \mathrm{Hypergeometric}(N, n, K)$, relabeling so the *small* number $K = 3$ plays the "choose from" role:

$$\mathbb{P}(X=1) = \frac{\binom{120}{1}\binom{380}{2}}{\binom{500}{3}} = \frac{120 \cdot 72010}{20708500} = \frac{8641200}{20708500} \approx \boxed{0.4173}$$

⋆ *Speed trick:* whichever of K, n is smaller, always relabel to put it in the top binomial coefficients — here it turns an infeasible calculation into three two-to-three-digit combinatorics terms.

Q7 — Pólya Urn with Stronger Reinforcement

An order-flow sentiment urn starts at 3 buy-signals, 2 sell-signals. Each draw, return the ball *plus 2 more* of the same color ($c = 2$ reinforcement). Find $\mathbb{P}$(Buy, Buy, Sell) in the first three draws.

Solution. Draw 1: $\mathbb{P}(\text{Buy}) = \frac{3}{5}$. Urn becomes $(3+2, 2) = (5, 2)$, total 7. Draw 2: $\mathbb{P}(\text{Buy} \mid \text{Buy}) = \frac{5}{7}$. Urn becomes $(5{+}2, 2) = (7, 2)$, total 9. Draw 3: $\mathbb{P}(\text{Sell} \mid \text{Buy,Buy}) = \frac{2}{9}$.

$$\mathbb{P}(\text{B,B,S}) = \frac{3}{5} \cdot \frac{5}{7} \cdot \frac{2}{9} = \frac{2}{21} \approx \boxed{0.0952}$$

⋆ *Intuition:* stronger reinforcement ($c = 2$ vs. $c = 1$) accelerates lock-in — the urn moves from $5:2$ to $7:2$ in just two draws, compounding the early buy-signal's influence faster than the standard $c = 1$ urn would.

Q8 — The Martingale Shortcut, Applied

A Pólya urn starts at 2 red, 3 black, reinforcement +1. Using *only* the martingale property (no recursive total-probability computation), state $\mathbb{P}$(3rd draw is red) and $\mathbb{E}[M_\infty]$.

Solution. The martingale property says $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for *every* n, including $n = \infty$:

$$\mathbb{P}(\text{3rd draw red}) = \mathbb{E}[M_2] = \mathbb{E}[M_0] = \frac{2}{2+3} = \boxed{0.4}$$

$$\mathbb{E}[M_\infty] = \frac{2}{2+3} = \boxed{0.4} \quad \text{(same number, by the same fact)}$$

⋆ *Speed trick: any* draw number in a Pólya urn has marginal probability equal to the initial proportion — this replaces an entire law-of-total-probability recursion (as in Example 12) with a single lookup, once you recognize the martingale property is in play.

Q9 — Lincoln–Petersen, Clean Numbers

To estimate active participants N in a dark pool, you tag $n_1 = 120$ orders Monday. Tuesday, $n_2 = 80$ sampled orders include $m_2 = 24$ Monday-tagged. Estimate N.

Solution.

$$\hat{N} = \frac{n_1 n_2}{m_2} = \frac{120 \cdot 80}{24} = \frac{9600}{24} = \boxed{400}$$

⋆ *Intuition:* $m_2/n_2 = 24/80 = 0.3$ estimates the population-wide tagged fraction n_1/N; solving $0.3 \approx 120/N$ gives the same 400 — the formula is just this cross-multiplication, memorized.

Q10 — Capstone: Chapman's Correction, Quantified

Same data as Q9 ($n_1 = 120, n_2 = 80, m_2 = 24$). Apply Chapman's bias-corrected estimator and state the direction and size of the correction versus Q9's raw $\hat{N}$.

Solution.

$$\hat{N}_C = \frac{(n_1+1)(n_2+1)}{m_2+1} - 1 = \frac{121 \cdot 81}{25} - 1 = \frac{9801}{25} - 1 = 392.04 - 1 = \boxed{391.04}$$

Versus Q9's $\hat{N} = 400$: Chapman's estimate is **lower** by ≈ 8.96, about 2.24%.

⋆ *Intuition:* raw $\hat{N} = n_1 n_2/m_2$ is biased *upward* (Jensen on $1/m_2$, Ch. 2.2), so the correction should always pull the estimate down — and it does here. With $m_2 = 24$ not tiny, the correction is modest ($\sim 2\%$); it would be far larger if m_2 were, say, 3 or 4, exactly where the raw estimator's instability is worst.

Drill order for review: re-derive Q4 (solve the FPC identity backwards), Q6 (duality trick from scratch), and Q10 (Chapman correction direction) closed-book. Together they test whether you can run the chapter's formulas in both directions, not just forward.